



UNIVERSITI PUTRA MALAYSIA

“BERILMU BERBAKTI”

CCS3700 – EMBEDDED SYSTEMS PROGRAMMING

LAB 5

MAKING MUSIC WITH PIEZO

LECTURER’S NAME:

DR. KHAIRUL AZHAR

STUDENT’S NAME:

MUHAMMAD ASLAM ROFI BIN MOHD RASU

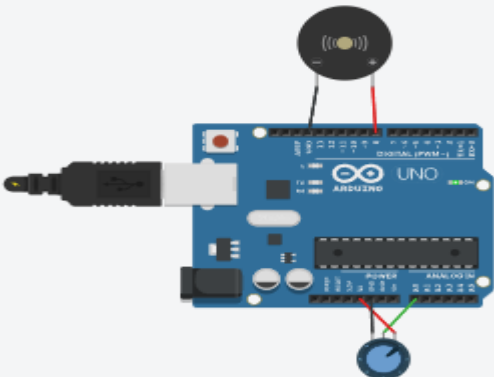
MATRIC NO:

230405

SUBMITTED DATE:

11/5/2026

## PART 1

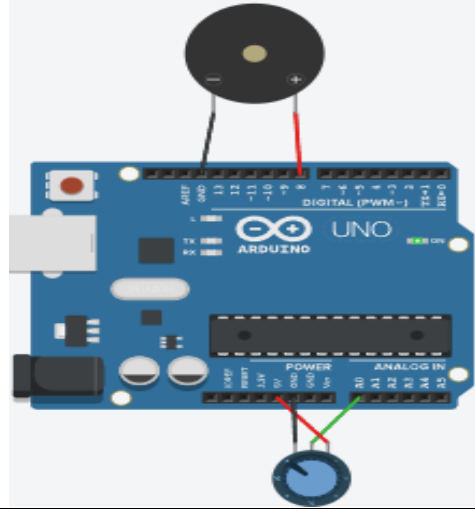
| PART 1 AND PART 2 LAYOUT                                                          | COMPONENTS                                                |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------|
|  | 1. ARDUINO UNO                                            |
|                                                                                   | 2. PIEZO BUZZER<br>- Neg = GND<br>- Pos = PIN 8           |
|                                                                                   | 3. POTENTIOMETER<br>- T1 = 5V<br>- T2 = GND<br>- MID = A0 |

| FULL PART 1 CODE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>// Define speaker pin int speakerPin = 8;  // TONES ===== ===== #define c 3830 // 261 Hz #define d 3400 // 294 Hz #define e 3038 // 329 Hz #define f 2864 // 349 Hz #define g 2550 // 392 Hz #define a 2272 // 440 Hz #define b 2028 // 493 Hz #define C 1912 // 523 Hz  // Melody data int length = 15; // the number of notes char notes[] = "ccggaagffeeddc"; // a space represents a rest int beats[] = { 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 1, 2, 4 }; int tempo = 300;  void setup() {   pinMode(speakerPin, OUTPUT); }</pre> | <pre>void loop() {   for (int i = 0; i &lt; length; i++) {     if (notes[i] == ' ') {       delay(beats[i] * tempo); // rest     } else {       playNote(notes[i], beats[i] * tempo);     }     // pause between notes     delay(tempo / 2);   }   delay(2000); // Wait 2 seconds before repeating the melody }  // Function to play a specific frequency (period) for a specific duration void playTone(int period, int duration) {   long elapsed_time = 0;   int pulse_width = period / 2; // 50% duty cycle</pre> | <pre>// duration is in milliseconds, convert to microseconds while (elapsed_time &lt; (duration * 1000L)) {   digitalWrite(speakerPin, HIGH);   delayMicroseconds(pulse_width);   digitalWrite(speakerPin, LOW);   delayMicroseconds(pulse_width);   elapsed_time += period; }  // Function to match the note character to its period void playNote(char note, int duration) {   char names[] = { 'c', 'd', 'e', 'f', 'g', 'a', 'b', 'C' };   int tones[] = { c, d, e, f, g, a, b, C };    for (int i = 0; i &lt; 8; i++) {     if (names[i] == note) {       playTone(tones[i], duration);     }   } }</pre> |

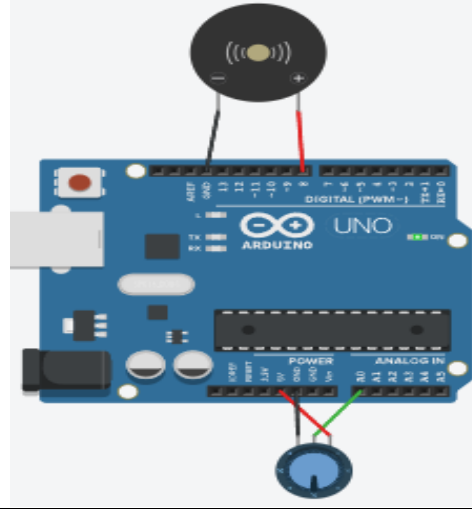
This program uses the frequencies provided in your snippet to play "Twinkle Twinkle Little Star". It pulses the speaker HIGH for half the period and LOW for half the period to create a 50% duty cycle, which generates the loudest and clearest tone.

## PART 2

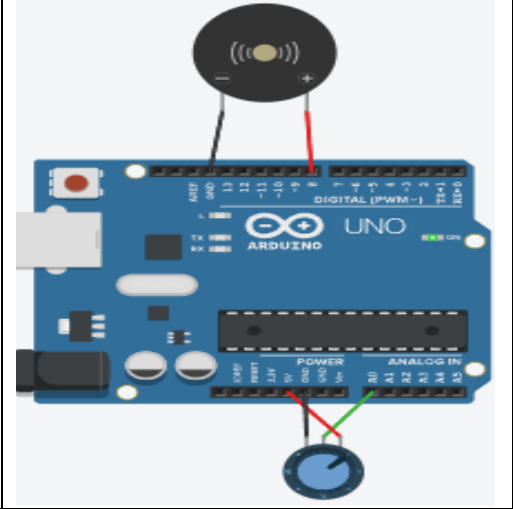
0 in Potentiometer



Half in potentiometer



Full in potentiometer



When the potentiometer is at 0, the buzzer simply makes no sound

When the potentiometer is at 512, the buzzer makes a sound but at a lower volume

When the potentiometer is at 1024, the buzzer makes a loud melody

This is possible due to the code:

When I use `map(volumeRead, 0, 1023, 0, period / 2)` in my code:

At **0% rotation** (0), the `pulse_width_high` is **0**, meaning the buzzer is silent.

At **50% rotation** (512), the `pulse_width_high` is **period / 4**, meaning the speaker is HIGH for 25% of the time and LOW for 75%.

At **100% rotation** (1023), `pulse_width_high` is **period / 2**, giving maximum vol (50% duty cycle).

### ENTIRE CODE OF PART 2

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>// Define pins int speakerPin = 8; int potPin = A0; // Potentiometer for volume control  // TONES #define c 3830 #define d 3400 #define e 3038 #define f 2864 #define g 2550 #define a 2272 #define b 2028 #define C 1912  // Melody data int length = 15; char notes[] = "ccggaagffeeddc "; int beats[] = { 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1, 2, 4 }; int tempo = 300;  void setup() {   pinMode(speakerPin, OUTPUT);   pinMode(potPin, INPUT); }</pre> | <pre>void loop() {   for (int i = 0; i &lt; length; i++) {     if (notes[i] == ' ') {       delay(beats[i] * tempo);     } else {       playNote(notes[i], beats[i] * tempo);     }     delay(tempo / 2);   }   delay(2000); }  void playNote(char note, int duration) {   long elapsed_time = 0;    // Read the potentiometer value (0 to 1023)   int volumeRead = analogRead(potPin);    // Map the volume to the HIGH pulse width.   // Max volume is period / 2 (50% duty cycle).   // Min volume is 0 (0% duty cycle).   long pulse_width_high = map(volumeRead, 0, 1023, 0, period / 2);   long pulse_width_low = period - pulse_width_high; // Ensure total period remains the same</pre> | <pre>while (elapsed_time &lt; (duration * 1000L)) {   if (pulse_width_high &gt; 0) {     digitalWrite(speakerPin, HIGH);     delayMicroseconds(pulse_width_high);   }    digitalWrite(speakerPin, LOW);   delayMicroseconds(pulse_width_low);    elapsed_time += period; }  void playNote(char note, int duration) {   char names[] = { 'c', 'd', 'e', 'f', 'g', 'a', 'b', 'C' };   int tones[] = { c, d, e, f, g, a, b, C };    for (int i = 0; i &lt; 8; i++) {     if (names[i] == note) {       playTone(tones[i], duration);     }   } }</pre> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|